

Experience Sampling Tool for Repetitive Skills Training in Sports using Voice User Interface

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ABSTRACT

Repetitive skills training (RST) is a commonly used method for improving skills. Although wearable devices and existing context-aware technologies allow us to easily detect objective data during RST, subjective data have not been collected effectively, even though both objective and subjective data are important for RST. In this paper, we propose and implement a prototype system, called MiQ, to collect subjective data with minimum workload during RST for sports. MiQ allows us to record subjective data hands-free via a voice user interface (VUI). We also discuss the future scope of the proposed prototype system.

CCS CONCEPTS

• **Computer systems organization** → **Embedded systems**; *Redundancy*; Robotics; • **Networks** → Network reliability.

KEYWORDS

Experience Sampling, Wearable Devices, Context-awareness, Voice User Interface

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1 INTRODUCTION

Repetitive skills training (RST) is a method for improving skills. Particularly in the sports domain, players who play any type and level of sports use RST to improve their skills. For example, shadow swings are a common training exercise for sports that involve hitting objects, such as baseball, tennis, and golf. The awareness and verbalization of subjective bodily senses (called metacognition in cognitive psychology) is an important factor in improving skills [3]. In addition to the objective performance data, most players observe their changes in feelings during periodically and use it for the next session. However, the pressure to take a note of their feeling is high because the time between cycles in an RST is short.

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Voice user interfaces (VUIs) are ubiquitous in our everyday life [1] and allow hands-free control of devices. The VUIs are implemented on off-the-shelf devices (e.g., smartphones, smartwatches, smart speakers, and headsets) and the opportunities for using it are increasing rapidly.

To the best of our knowledge, in the field of sports, most of the existing research tackles the collection and feedback of objective motion data during an RST [5, 6]. However, subjective data is also important for enriching RST. Research on collecting subjective information during an RST is a challenging research topic.

In this paper, we propose a system called *MiQ* for collecting subjective physical information during an RST using a VUI. The system detects the steps during RST using a wearable device, and interrupts the steps via wireless headphones to provide encouraging inputs, such as “the feeling of the last attempt” and “the aim of the next attempt” at a particular moment.

2 RELATED WORKS

In the field of sports, real-time audio feedback is used for providing objective data [5, 6]. For example, Ohnishi et al. [5] proposed a sound feedback system for equalizing the foot pressure balance of runners between their left and right feet. In contrast, Katsura [2] explored the effects of cheering on the performance of rock climbers using machine-generated voice and computer graphics. These studies used audio interfaces during sports training but did not attempt to collect subjective players’ subjective feelings during RST.

VUIs allow us to control a device hands-free. Several studies have explored the applicability of VUIs to a labeling and experience sampling tool [4, 7]. Mairitha et al. [4] proposed a dialogue-based annotation system for activity recognition. A user can make an annotation using the voice agent on the system, but the user needs to execute a trigger to start the annotation process. Wei et al. [7] implemented a voice-based experience sampling application on a smart speaker. The application created questions using a smart speaker based on the context around the speaker. However, the application was designed for a home usage scenario. VUIs generate new opportunities for hands-free sampling but have not been used for expanding RST.

3 VUI-BASED EXPERIENCE SAMPLING FOR SPORTS

In this paper, we propose a hands-free subjective data collection system, called *MiQ*, for RST. As an example of RST, we applied *MiQ* for bat swing training in baseball.

Figure 1 shows the concept of VUI-based experience sampling. A cycle of an RST comprises three steps: (1) set the aim of the attempt, (2) perform the attempt, and (3) reflect on the result of the attempt,

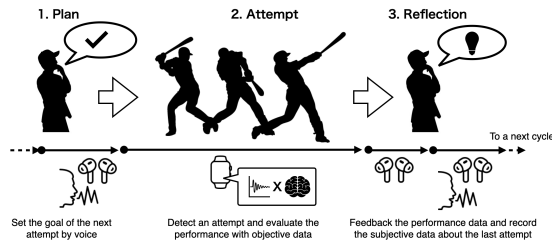


Figure 1: Concept of VUI-based experience sampling

including subjective feeling and objective information (e.g., swing speed) for the next RST cycle. A player repeats the process multiple times to adjust their real physical motions and feelings.

3.1 System Design

To record the subjective data during an RST, a player needs to record it hands-free between each session and save the data as text. To fulfill this requirement, MiQ uses three steps, i.e., plan, attempt, and reflect, which also are phases of the RST, and measures the performance of the attempt automatically. In the case of a bat swing, MiQ detects the swing speed. At the end of the swing event, MiQ automatically provides the performance as feedback via a headset and starts a voice recognition process automatically. The player records their physical feeling during the attempt. Subsequently, MiQ suggests declaring the aim of the next attempt. The player can check the recorded subjective and objective data on a smartphone application after the RST.

3.2 Prototype

A prototype of MiQ was constructed with three devices, an iPhone (iOS 14.5), an Apple Watch (watchOS 7.4.1), and a bone conduction headphone (as a headset), as shown in Figure 2.

(a) The watchOS application detects a phase of the RST on the device using motion sensors. The application collects user-accelerometer and rotation data at a frequency of 100 Hz and detects motion events every second using a neural network-based machine learning (ML) model on the Apple Watch. We created the ML model using CreateML on Xcode. The dataset of bat swings was collected from five baseball players with 20 bat swings each.

(b) When the watch application detects a swing event, the application notifies the detected phase to the smartphone application. Based on the trigger, the smartphone application sends a voice message to the (“How was the last attempt?”) via a headset. The player makes an observation about the attempt using voice. The smartphone application converts the audio data to text using a speech-to-text application programming interface on iOS.

(c) When the player responds to the first question, the smartphone application asks the question about the aim of the next attempt (“What is the aim of the next attempt?”). After the answer, the player performs the next attempt.

During an RST session, a player continuously performs this process.

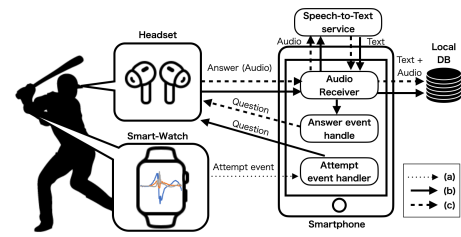


Figure 2: System architecture of MiQ

4 FUTURE WORKS

In the future work, we aim to evaluate our proposed system with sports players in multiple conditions. In our proposed system, the user experience (UX) would change depending on the interrupt timing because the system prompts the user for inputs in the middle of repetitive practice. Therefore, we investigate the configuration that maximizes the user experience by controlling the interrupt timing and its frequency. In addition to the sports field, RST is applied to various fields, such as music, art, and surgery. We also aim to investigate the applicability of RST in these domains.

5 CONCLUSION

In this paper, we proposed the concept of a VUI-based experience sampling method for RST, and designed and implemented a prototype of the concept, called MiQ. The prototype allowed us to collect subjective data (such as physical feeling and aim of the RST) of a player hands- and trigger-free. In addition, we discussed the future scope of this research based on the prototype.

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